Chandubhai S Patel Institute of Technology

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

DEPSTAR

Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement: Asynchronous Processing with Event-Driven Architecture.

Screenshots (Supplementary Problems are also implemented):

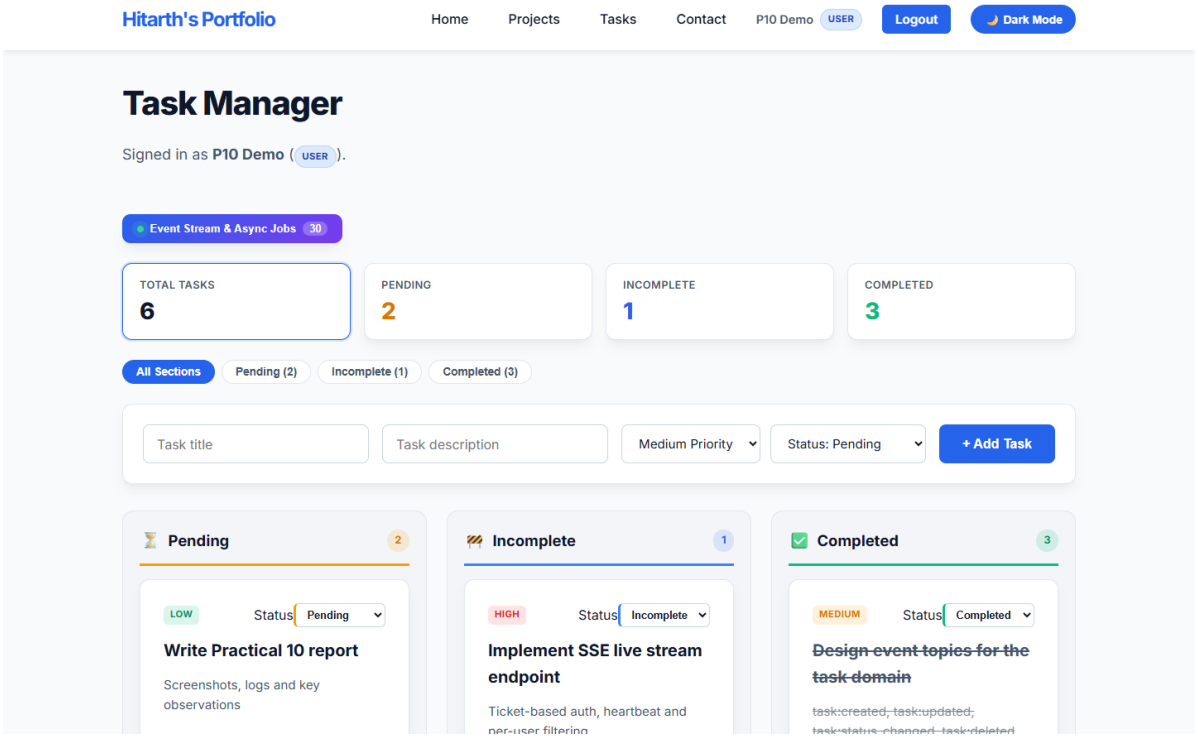


Figure 1: *Tasks page connected to the Node.js + MongoDB backend, with the new "Event Stream & Async Jobs" button (live event counter) that opens the event-driven architecture panel.*

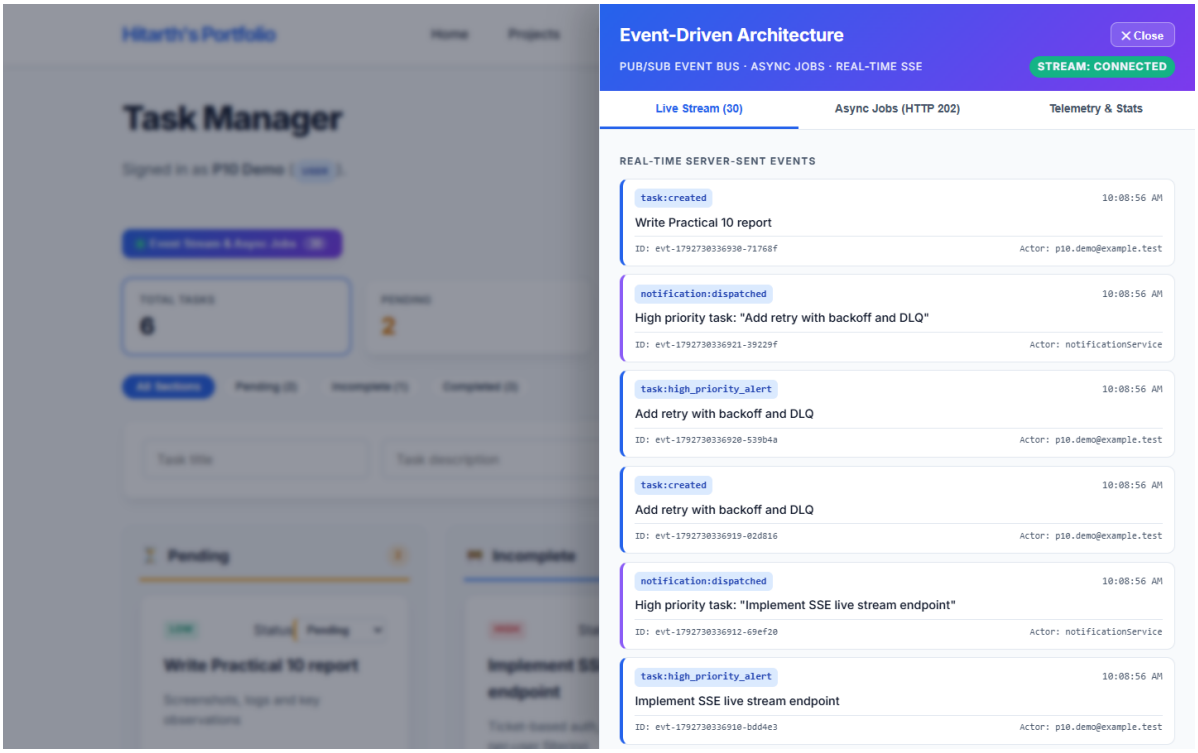


Figure 2: *Event-Driven Architecture drawer on the Live Stream tab. The SSE stream is CONNECTED (opened with a single-use ticket, so the JWT never appears in the URL) and recent events are loaded from history.*

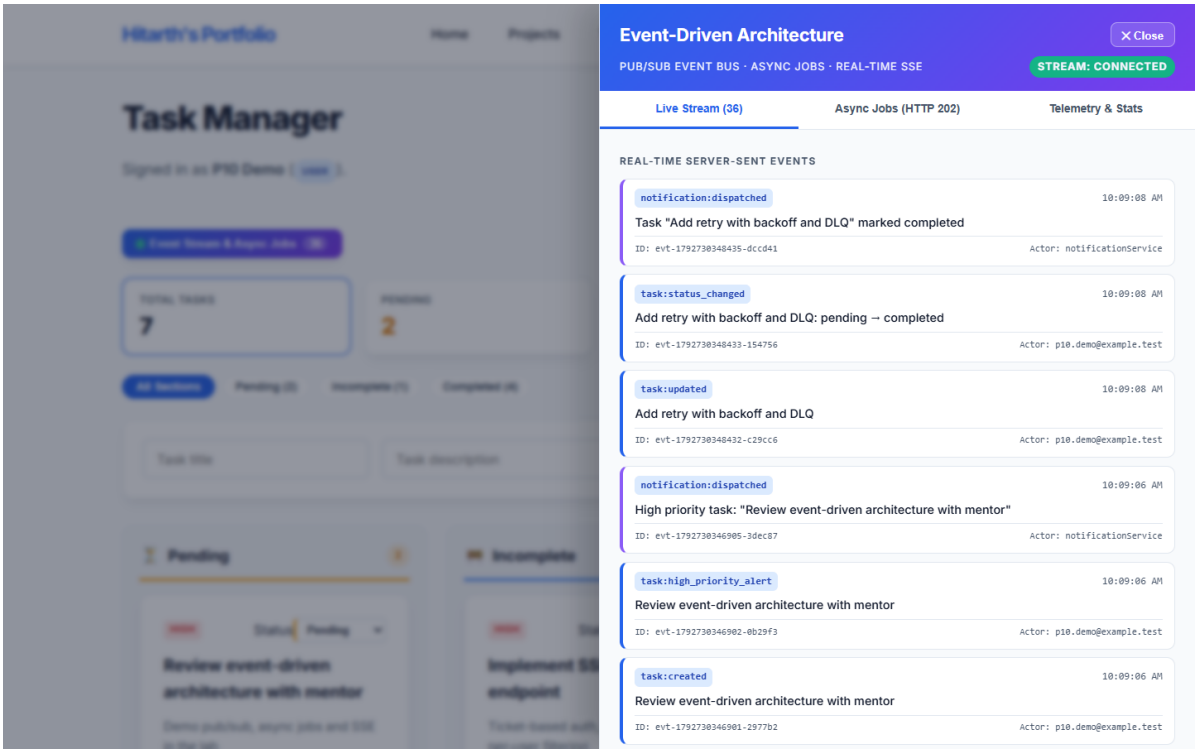


Figure 3: *Creating a high-priority task and completing another from the dashboard publishes task:created, task:high_priority_alert, task:updated and task:status_changed; the notification subscriber reacts with notification:dispatched, all pushed to the browser in real time.*

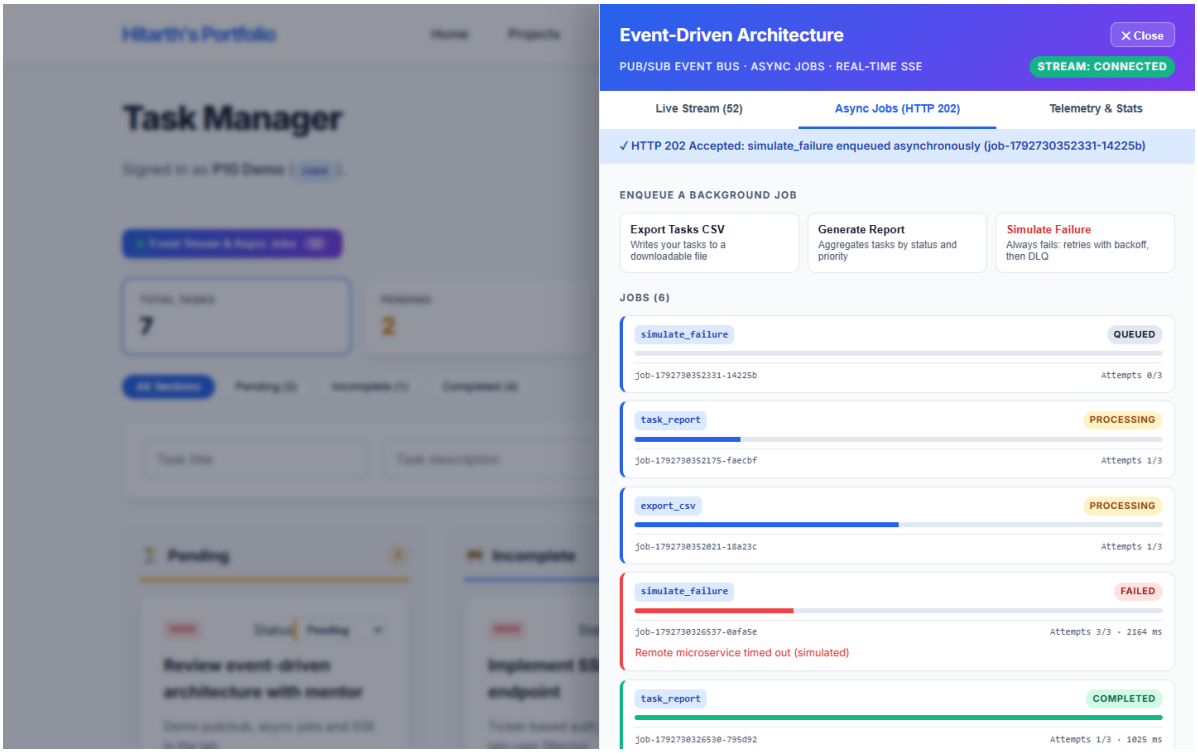


Figure 4: *Async Jobs tab: each job request returns HTTP 202 Accepted immediately and is processed by background workers (concurrency 2), with live progress bars driven by job:progress events.*

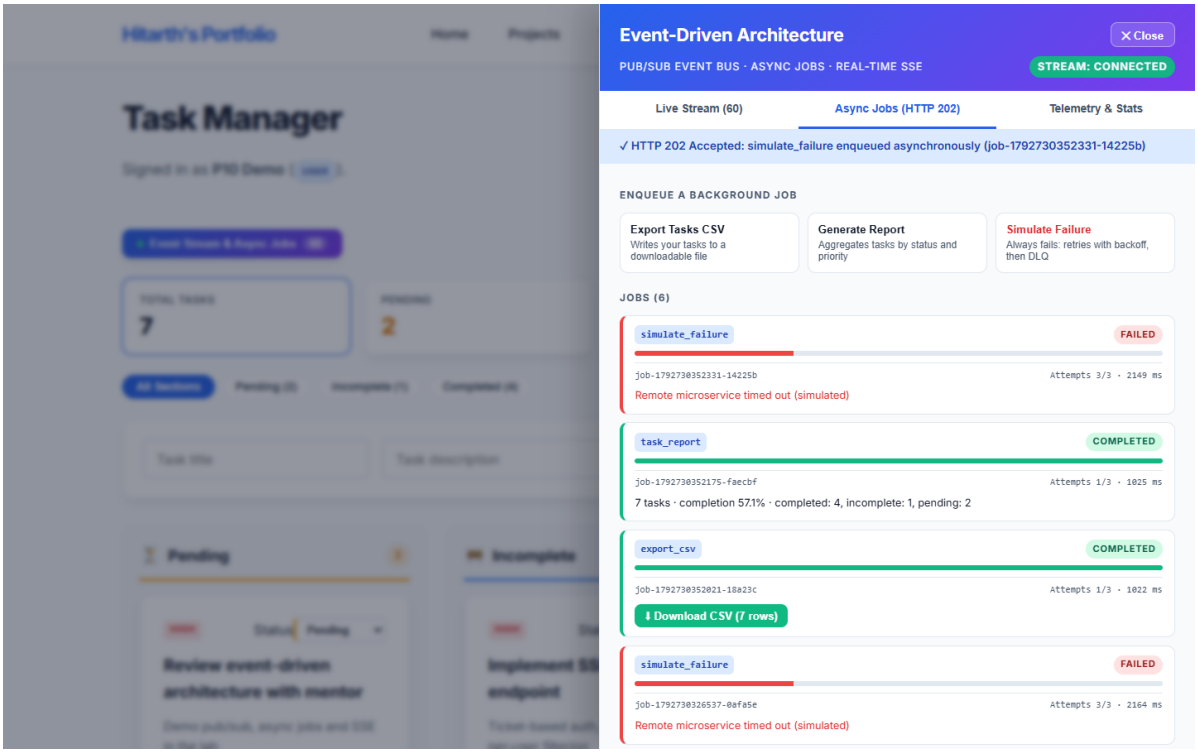


Figure 5: *Completed jobs: export_csv produced a downloadable CSV, task_report returned an aggregate summary, and simulate_failure FAILED after 3/3 attempts (exponential backoff) and was moved to the Dead Letter Queue.*

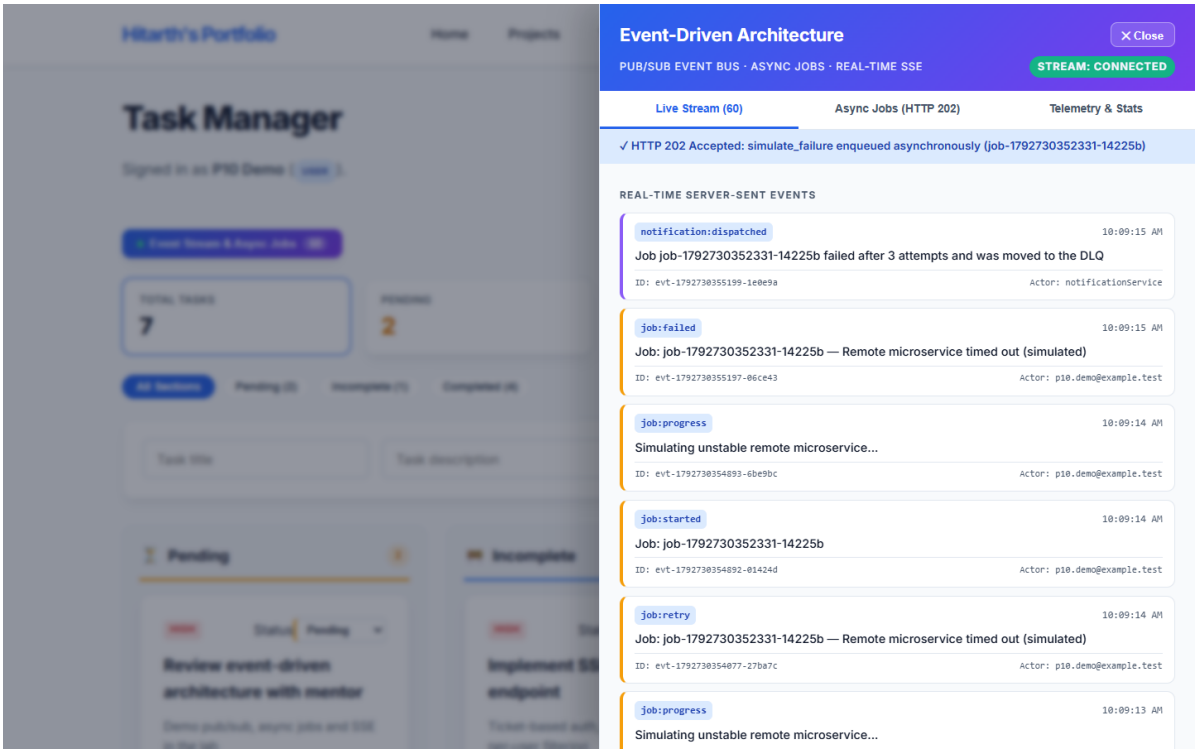


Figure 6: *Live Stream showing the job lifecycle events (job:enqueued, job:started, job:progress, job:retry, job:completed, job:failed) with event IDs, timestamps and actors.*

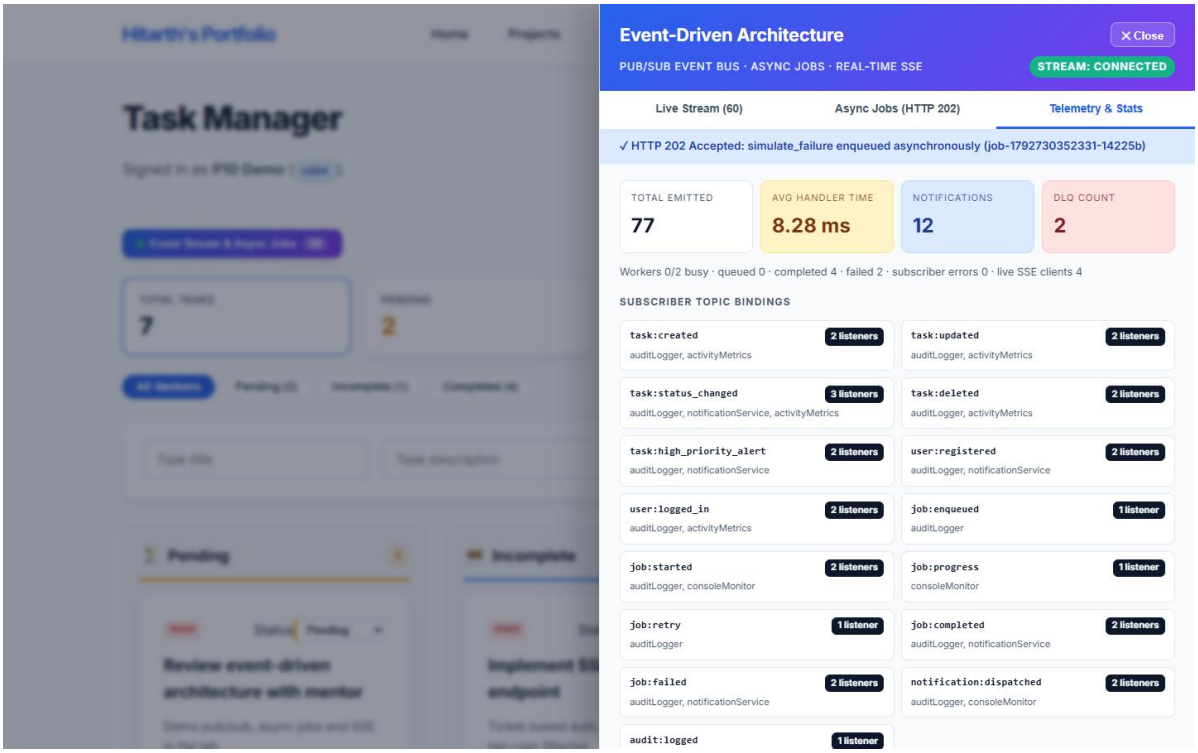


Figure 7: *Telemetry & Stats tab: total events emitted, average subscriber handler time, notifications sent, DLQ count, worker/queue status, and the subscribers bound to each event topic.*

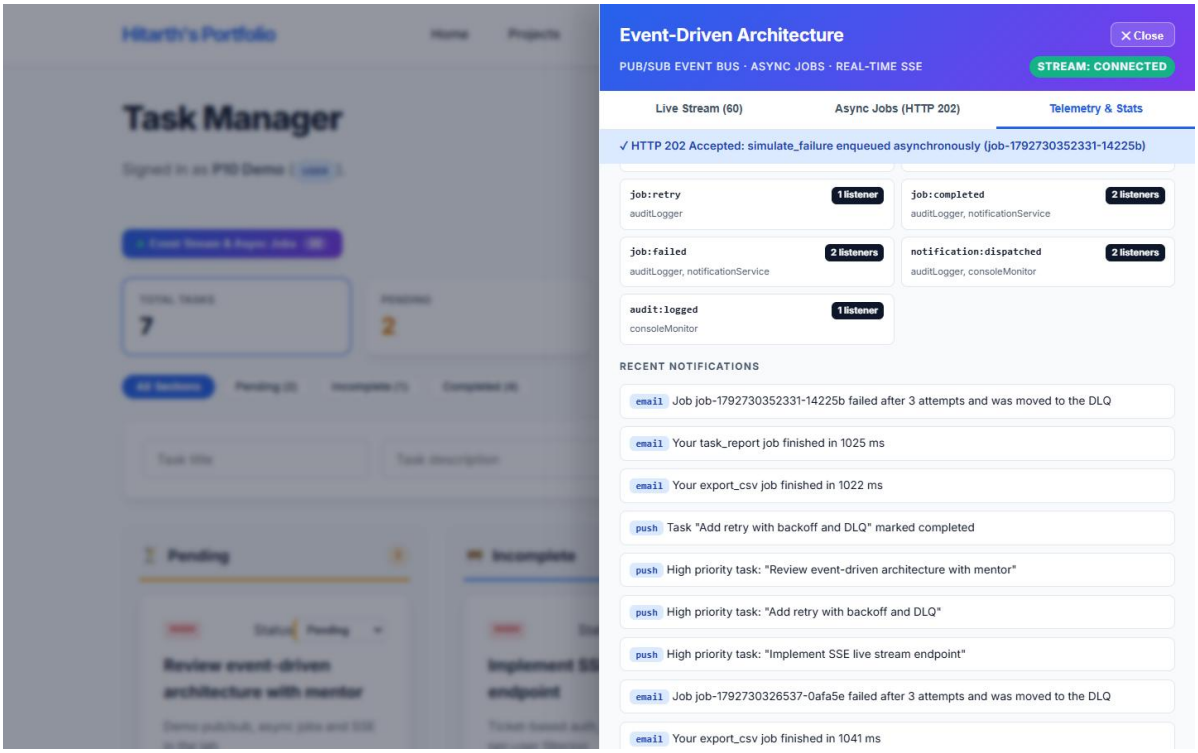


Figure 8: *Telemetry tab scrolled: remaining topic bindings and the recent notifications produced by the notification subscriber (job finished, job failed / moved to DLQ, high-priority alerts).*

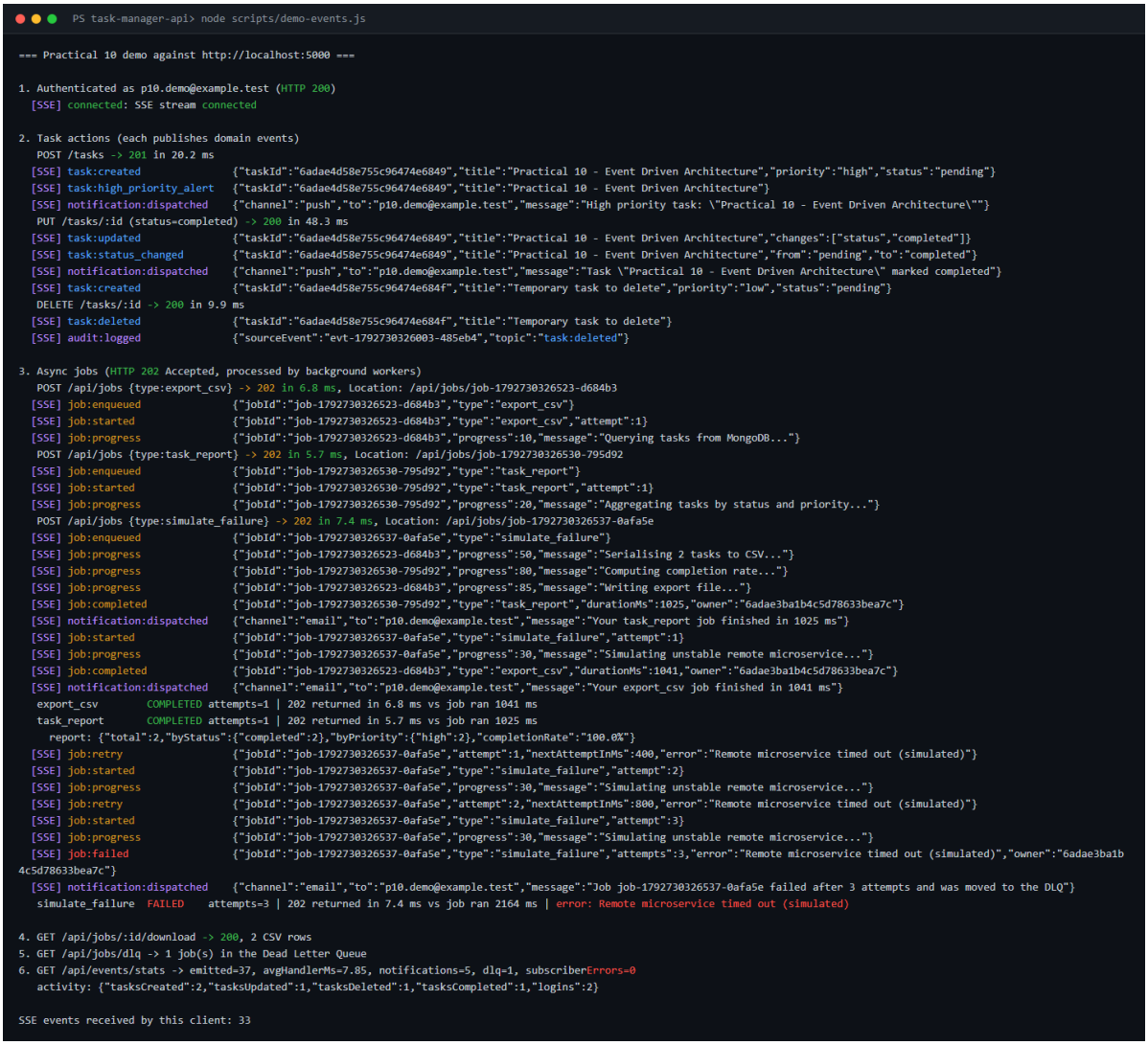


Figure 9: *Terminal log of scripts/demo-events.js: the client receives every event over SSE; POST /api/jobs answers 202 in ~5-7 ms while the jobs themselves run ~1-2 s, proving the request is non-blocking.*

logs/events.csv (audit subscriber output)


```
timestamp,event_id,topic,actor,payload
2026-10-23T04:38:56.920Z,evt-1792730336920-539b4a,task:high_priority_alert,p10.demo@example.test,"{""taskId"":""6adae4e08e755c96474e685a"",""title"":""Add retry with backoff and DLQ""}"
2026-10-23T04:38:56.921Z,evt-1792730336921-39229f,notification:dispatched,notificationService,"{""channel"":""push"",""to"":""p10.demo@example.test"",""message"":""High priority task: \""Add retry with backoff and DLQ\""""}"
2026-10-23T04:38:56.930Z,evt-1792730336930-71768f,task:created,p10.demo@example.test,"{""taskId"":""6adae4e08e755c96474e685c"",""title"":""Write Practical 10 report"",""priority"":""low"",""status"":""pending""}"
2026-10-23T04:39:06.901Z,evt-1792730346901-2977b2,task:created,p10.demo@example.test,"{""taskId"":""6adae4ea8e755c96474e6861"",""title"":""Review event-driven architecture with mentor"",""priority"":""high"",""status"":""pending""}"
2026-10-23T04:39:06.902Z,evt-1792730346902-0b29f3,task:high_priority_alert,p10.demo@example.test,"{""taskId"":""6adae4ea8e755c96474e6861"",""title"":""Review event-driven architecture with mentor""}"
2026-10-23T04:39:06.905Z,evt-1792730346905-3dec87,notification:dispatched,notificationService,"{""channel"":""push"",""to"":""p10.demo@example.test"",""message"":""High priority task: \""Review event-driven architecture with mentor\""""}"
2026-10-23T04:39:08.433Z,evt-1792730348433-154756,task:status_changed,p10.demo@example.test,"{""taskId"":""6adae4e08e755c96474e685a"",""title"":""Add retry with backoff and DLQ"",""from"":""pending"",""to"":""completed""}"
2026-10-23T04:39:08.432Z,evt-1792730348432-c29cc6,task:updated,p10.demo@example.test,"{""taskId"":""6adae4e08e755c96474e685a"",""title"":""Add retry with backoff and DLQ"",""changes"":[""status"",""completed""]}"
2026-10-23T04:39:08.435Z,evt-1792730348435-dccd41,notification:dispatched,notificationService,"{""channel"":""push"",""to"":""p10.demo@example.test"",""message"":""Task \""Add retry with backoff and DLQ\"" marked completed""}"
2026-10-23T04:39:12.021Z,evt-1792730352021-75d083,job:enqueued,p10.demo@example.test,"{""jobId"":""job-1792730352021-18a23c"",""type"":""export_csv""}"
2026-10-23T04:39:12.023Z,evt-1792730352023-0ea296,job:started,p10.demo@example.test,"{""jobId"":""job-1792730352021-18a23c"",""type"":""export_csv"",""attempt"":1}"
2026-10-23T04:39:12.176Z,evt-1792730352176-86f78f,job:enqueued,p10.demo@example.test,"{""jobId"":""job-1792730352175-faecbf"",""type"":""task_report""}"
2026-10-23T04:39:12.178Z,evt-1792730352178-166985,job:started,p10.demo@example.test,"{""jobId"":""job-1792730352175-faecbf"",""type"":""task_report"",""attempt"":1}"
2026-10-23T04:39:12.331Z,evt-1792730352331-b248c1,job:enqueued,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""type"":""simulate_failure""}"
2026-10-23T04:39:13.045Z,evt-1792730353045-dc2b52,job:completed,p10.demo@example.test,"{""jobId"":""job-1792730352021-18a23c"",""type"":""export_csv"",""durationMs"":1022,""owner"":""6adae3ba1b4c5d78633bea7c""}"
2026-10-23T04:39:13.047Z,evt-1792730353047-18b71f,notification:dispatched,notificationService,"{""channel"":""email"",""to"":""p10.demo@example.test"",""message"":""Your export_csv job finished in 1022 ms""}"
2026-10-23T04:39:13.048Z,evt-1792730353048-2bc65c,job:started,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""type"":""simulate_failure"",""attempt"":1}"
2026-10-23T04:39:13.204Z,evt-1792730353203-36752a,job:completed,p10.demo@example.test,"{""jobId"":""job-1792730352175-faecbf"",""type"":""task_report"",""durationMs"":1025,""owner"":""6adae3ba1b4c5d78633bea7c""}"
2026-10-23T04:39:13.206Z,evt-1792730353206-f2b2ed,notification:dispatched,notificationService,"{""channel"":""email"",""to"":""p10.demo@example.test"",""message"":""Your task_report job finished in 1025 ms""}"
2026-10-23T04:39:13.361Z,evt-1792730353361-cbeede,job:retry,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""attempt"":1,""nextAttemptInMs"":400,""error"":""Remote microservice timed out (simulated)""}"
2026-10-23T04:39:13.771Z,evt-1792730353771-33df3b,job:started,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""type"":""simulate_failure"",""attempt"":2}"
2026-10-23T04:39:14.078Z,evt-1792730354077-27ba7c,job:retry,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""attempt"":2,""nextAttemptInMs"":800,""error"":""Remote microservice timed out (simulated)""}"
2026-10-23T04:39:14.893Z,evt-1792730354892-01424d,job:started,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""type"":""simulate_failure"",""attempt"":3}"
2026-10-23T04:39:15.197Z,evt-1792730355197-06ce43,job:failed,p10.demo@example.test,"{""jobId"":""job-1792730352331-14225b"",""type"":""simulate_failure"",""attempts"":3,""error"":""Remote microservice timed out (simulated)"",""owner"":""6adae3ba1b4c5d78633bea7c""}"
2026-10-23T04:39:15.199Z,evt-1792730355199-1e0e9a,notification:dispatched,notificationService,"{""channel"":""email"",""to"":""p10.demo@example.test"",""message"":""Job job-1792730352331-14225b failed after 3 attempts and was moved to the DLQ""}"
2026-10-23T04:41:24.226Z,evt-1792730484226-4ce7fc,user:logged_in,p10.demo@example.test,"{""email"":""p10.demo@example.test"",""role"":""user""}"
```


Figure 10: *logs/events.csv written by the audit-logger subscriber: one row per domain event with timestamp, event ID, topic, actor and JSON payload.*

logs/server.log (event bus + subscriber console output)


```
[EventBus] 2026-10-23T04:39:12.528Z job:progress evt-1792730352528-64fd9f actor=p10.demo@example.test
[Monitor] job:progress {"jobId":"job-1792730352021-18a23c","progress":50,"message":"Serialising 7 tasks to CSV..."}
[EventBus] 2026-10-23T04:39:12.890Z job:progress evt-1792730352890-3a8f1c actor=p10.demo@example.test
[Monitor] job:progress {"jobId":"job-1792730352175-faecbf","progress":80,"message":"Computing completion rate..."}
[EventBus] 2026-10-23T04:39:13.031Z job:progress evt-1792730353031-3ba0eb actor=p10.demo@example.test
[Monitor] job:progress {"jobId":"job-1792730352021-18a23c","progress":85,"message":"Writing export file..."}
[EventBus] 2026-10-23T04:39:13.045Z job:completed evt-1792730353045-dc2b52 actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:13.047Z notification:dispatched evt-1792730353047-18b71f actor=notificationService
[EventBus] 2026-10-23T04:39:13.048Z job:started evt-1792730353048-2bc65c actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:13.048Z job:progress evt-1792730353048-1ee497 actor=p10.demo@example.test
[Monitor] notification:dispatched {"channel":"email","to":"p10.demo@example.test","message":"Your export_csv job finished in 1022 ms"}
[Monitor] job:started {"jobId":"job-1792730352331-14225b","type":"simulate_failure","attempt":1}
[Monitor] job:progress {"jobId":"job-1792730352331-14225b","progress":30,"message":"Simulating unstable remote microservice..."}
[EventBus] 2026-10-23T04:39:13.204Z job:completed evt-1792730353203-36752a actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:13.206Z notification:dispatched evt-1792730353206-f2b2ed actor=notificationService
[Monitor] notification:dispatched {"channel":"email","to":"p10.demo@example.test","message":"Your task_report job finished in 1025 ms"}
[EventBus] 2026-10-23T04:39:13.361Z job:retry evt-1792730353361-cbeede actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:13.771Z job:started evt-1792730353771-33df3b actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:13.771Z job:progress evt-1792730353771-542e78 actor=p10.demo@example.test
[Monitor] job:started {"jobId":"job-1792730352331-14225b","type":"simulate_failure","attempt":2}
[Monitor] job:progress {"jobId":"job-1792730352331-14225b","progress":30,"message":"Simulating unstable remote microservice..."}
[EventBus] 2026-10-23T04:39:14.078Z job:retry evt-1792730354077-27ba7c actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:14.893Z job:started evt-1792730354892-01424d actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:14.893Z job:progress evt-1792730354893-6be9bc actor=p10.demo@example.test
[Monitor] job:started {"jobId":"job-1792730352331-14225b","type":"simulate_failure","attempt":3}
[Monitor] job:progress {"jobId":"job-1792730352331-14225b","progress":30,"message":"Simulating unstable remote microservice..."}
[EventBus] 2026-10-23T04:39:15.197Z job:failed evt-1792730355197-06ce43 actor=p10.demo@example.test
[EventBus] 2026-10-23T04:39:15.199Z notification:dispatched evt-1792730355199-1e0e9a actor=notificationService
[Monitor] notification:dispatched {"channel":"email","to":"p10.demo@example.test","message":"Job job-1792730352331-14225b failed after 3 attempts and was moved to the DLQ"}
[SSE] client disconnected: p10.demo@example.test
[SSE] client disconnected: p10.demo@example.test
[EventBus] 2026-10-23T04:41:24.226Z user:logged_in evt-1792730484226-4ce7fc actor=p10.demo@example.test
[EventBus] 2026-10-23T04:41:24.237Z audit:logged evt-1792730484237-e346da actor=auditLogger
[Monitor] audit:logged {"sourceEvent":"evt-1792730484226-4ce7fc","topic":"user:logged_in"}
```


Figure 11: *logs/server.log: [EventBus] publish lines and [Monitor] subscriber output, showing the retry with backoff and the final job:failed event followed by the failure notification.*



Figure 12: *logs/notifications.log written by the notification subscriber (simulated email / push channels), fully decoupled from the controllers that produced the events.*

Github Repo Link: https://github.com/hitarth1812/Student-Portfolio